



Stakeholder Engagement

Optimizing RAG Document Retrieval for Agronomic Advice

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1. Selected Stakeholder Groups

Based on our stakeholder map, we select the two groups that span the full delivery of our retrieval system:

- **Smallholder Farmers:** The end users whose natural-language questions drive the system and whose livelihoods depend on the quality of the retrieval advice.
- **Agricultural Extension Agents:** The field intermediaries who bridge the gap between technical knowledge and farmer practice, and whose adoption decisions determine whether the system reaches the intended communities.

Together, these groups cover the *demand side* (farmers formulating real queries) and the *delivery side* (agents vetting and channelling retrieved information). Engaging both ensures that the retrieval engine is shaped by the people closest to the problem.

2. What Matters Most to Each Group

Smallholder farmers care most about the *accuracy and safety* of the advice they receive: a wrong pesticide recommendation or an off-zone fertiliser regime can cause crop loss and financial hardship. Accessibility also ranks high—queries must be understood even when phrased in local languages or mixed codes, and answers must be usable by people with limited literacy or connectivity.

Extension agents prioritise *efficiency and professional credibility*. They need to find the right factsheet quickly during field visits, and they need the tool to reinforce—not undermine—their expertise. Coverage matters too: if the knowledge base has blind spots on common local crops, the tool loses trust.

Both groups share a concern for *trustworthiness*: the system should surface expert-validated content, not commercial advertising or outdated research.

3. Potential Harms and Benefits

4. Engagement Methods from the Action Catalogue

With regard to the Engage2020 Action Catalogue (actioncatalogue.eu) and filtering for methods suited to participatory research with affected communities, small group sizes, and low-resource contexts, we selected the following methods:

	Potential Benefits	Potential Harms
Smallholder Farmers	Faster, more equitable access to expert-validated agronomic advice regardless of proximity to an extension office or internet connectivity. Empowerment through information: better-informed decisions on pest management, soil health, and climate adaptation. Reduction in the information gap between well-connected and marginalised rural communities.	Retrieval of advice designed for a different agroecological zone (wrong pesticide, wrong planting window) leading to crop damage or financial loss. Exclusion of under-represented crops or regions from the corpus, systematically under-serving certain farming communities. Over-reliance on the tool without professional cross-checking, creating a false sense of certainty.
Extension Agents	Efficiency gains during field visits: instant, ranked access to the right factsheet instead of manual search through PDF repositories. Extended reach: agents can cover more farmers with higher-quality, evidence-based advice. Professional development through exposure to a broader knowledge base.	Perception that the tool threatens their professional role, leading to resistance or disengagement. Misrepresentation of complex agronomic advice when documents are retrieved out of context. Additional cognitive burden if the interface is poorly designed for low-resource field conditions (offline, low-bandwidth, small screens).

- **For smallholder farmers — Participatory Design (co-design workshops):** Farmers contribute real queries and evaluate retrieval outputs face-to-face, ensuring the system reflects actual information needs.
- **For extension agents — User Committee (standing advisory panel):** A small, recurring committee of agents provides structured, longitudinal feedback that directly shapes iteration cycles.

5. Why Each Method Fits

5a. Participatory Design for Smallholder Farmers

Participatory Design (PD) is a methodology rooted in the Scandinavian tradition of democratic co-creation, where the people most affected by a technology participate directly in shaping it. Rather than treating end-users as passive subjects of testing, PD positions them as experts of their own lived experience—an epistemological stance that is particularly important when building information systems for communities whose knowledge, language, and farming context are poorly represented in formal academic literature. The Engage2020 Action Catalogue lists PD as well-suited for projects aiming to surface ideas for new technology development and to involve affected communities in shaping research agendas.

For our project, PD takes the form of **co-design workshops** conducted with small groups of smallholder farmers (8–12 participants per session) in their local communities. During these

workshops, farmers are presented with a set of agricultural queries—some drawn from the competition training set, others formulated by the farmers themselves in real time—and shown the top-5 documents that our retrieval system returns. Farmers then evaluate, in their own terms, whether the retrieved factsheets actually answer the question they had in mind.

This is fundamentally different from asking an external expert to judge relevance: it captures the farmer’s understanding of what constitutes a useful answer, which may diverge from academic judgement. For instance, a factsheet on maize nutrient deficiency might score highly on expert relevance labels, but a farmer whose maize variety is specific to her altitude and soil type may consider it irrelevant—a discrepancy that only emerges through direct participation.

This method fits our stakeholder group for several structural reasons:

1. Smallholder farmers in Sub-Saharan Africa often operate in oral, communal decision-making cultures where knowledge is shared through dialogue rather than written forms. PD workshops, especially when facilitated in local languages, align naturally with this communication mode, lowering the barrier to meaningful participation.
2. PD does not require participants to be literate or digitally fluent. Farmers interact with printed or projected retrieval results, and facilitators read or summarise content aloud when needed. Visual aids such as crop photographs or colour-coded relevance cards can replace text-heavy evaluation forms.
3. PD generates qualitative data that no automated evaluation metric can capture: *why* a document feels trustworthy or misleading, what contextual information is missing, and what follow-up questions the farmer would ask. These insights feed directly into our retrieval pipeline—for example, by identifying hard negatives that should be penalised during re-ranking, or by surfacing crop-topic gaps that signal corpus-level bias.

A practical consideration is cost. PD workshops require physical facilitation, travel to farming communities, and potentially compensation for participants’ time. However, the method scales well when embedded within existing extension-service visits or farmer cooperative meetings, minimising logistical overhead. The workshops also yield a secondary benefit: they build trust and ownership, making the farmer community more likely to adopt and champion the tool when it reaches deployment.

The key risk is *tokenism*—running workshops that collect input but never act on it. We mitigate this by committing to a feedback loop: after each workshop round, we document which retrieval failures were flagged, implement re-ranking adjustments, and share the results in a follow-up session.

5b. User Committee for Agricultural Extension Agents

A User Committee, as catalogued in the Engage2020 Action Catalogue, is a standing advisory group composed of representative stakeholders who meet regularly over the life of a project. Unlike one-off consultations, a User Committee provides sustained, structured engagement. Members develop familiarity with the system’s evolution and can offer increasingly informed and nuanced feedback over successive iterations. The Action Catalogue identifies this method as particularly effective when the project needs ongoing input from a stakeholder group with moderate-to-high influence—matching the profile of extension agents in our stakeholder mapping (“Manage Closely” quadrant: high interest, high influence).

We propose a User Committee of **5–8 agricultural extension agents** recruited from different agroecological zones covered by the corpus (e.g., humid forest, guinea savanna, highland). The committee convenes bi-weekly, alternating between in-person meetings and lightweight remote check-ins via phone or messaging platforms such as WhatsApp. Each session follows a structured

protocol: agents review a batch of recent retrieval outputs, annotate cases where the ranking failed or succeeded, and flag emerging query patterns from their field interactions. Over time, the committee builds a shared annotation corpus that serves as a qualitative complement to the quantitative nDCG@5 metric.

The fit between User Committees and extension agents is grounded in several characteristics:

1. Extension agents are trained professionals with domain expertise. They possess the agronomic vocabulary and critical judgement needed to evaluate whether a retrieved factsheet is not just topically related but practically sound for a given farmer context.
2. Agents occupy a gatekeeping position in the information delivery chain. A User Committee gives them a formal role in shaping the tool, which directly addresses the engagement risk identified in our stakeholder mapping: the perception that the system threatens their professional autonomy.
3. The longitudinal nature of the method is critical. Our retrieval pipeline evolves as we experiment with different approaches, and each configuration change may introduce new failure modes. A standing committee can track these shifts over time.
4. Extension agents are geographically distributed across the regions the corpus covers. A committee drawn from multiple zones provides a natural mechanism for detecting regional bias in retrieval performance.

The practical feasibility of this method is high. Extension agents are already embedded in institutional structures (government extension services, NGO programmes) that facilitate recruitment and continuity. The main challenge is attrition: committee members may be reassigned or lose motivation. We address this by keeping the committee small, recognising contributions publicly, and ensuring that every round of feedback leads to a visible system improvement.

6. Influence on Dataset and Model Design

Both engagement methods feed directly into our retrieval pipeline. Participatory Design workshops with farmers generate new, real-world queries that expose corpus coverage gaps—crops, pests, or agroecological conditions that the existing 695 factsheets do not adequately address—prompting targeted corpus expansion or the introduction of query-rewriting strategies to handle under-represented topics.

The User Committee’s ongoing annotations create a supplementary set of expert relevance judgements that can be used to fine-tune re-ranking models beyond the competition-provided qrels, particularly for hard-negative cases.

Together, these feedback streams ensure that data sourcing, relevance labelling, and ranking model design are iteratively grounded in the needs and knowledge of the communities the system is meant to serve.